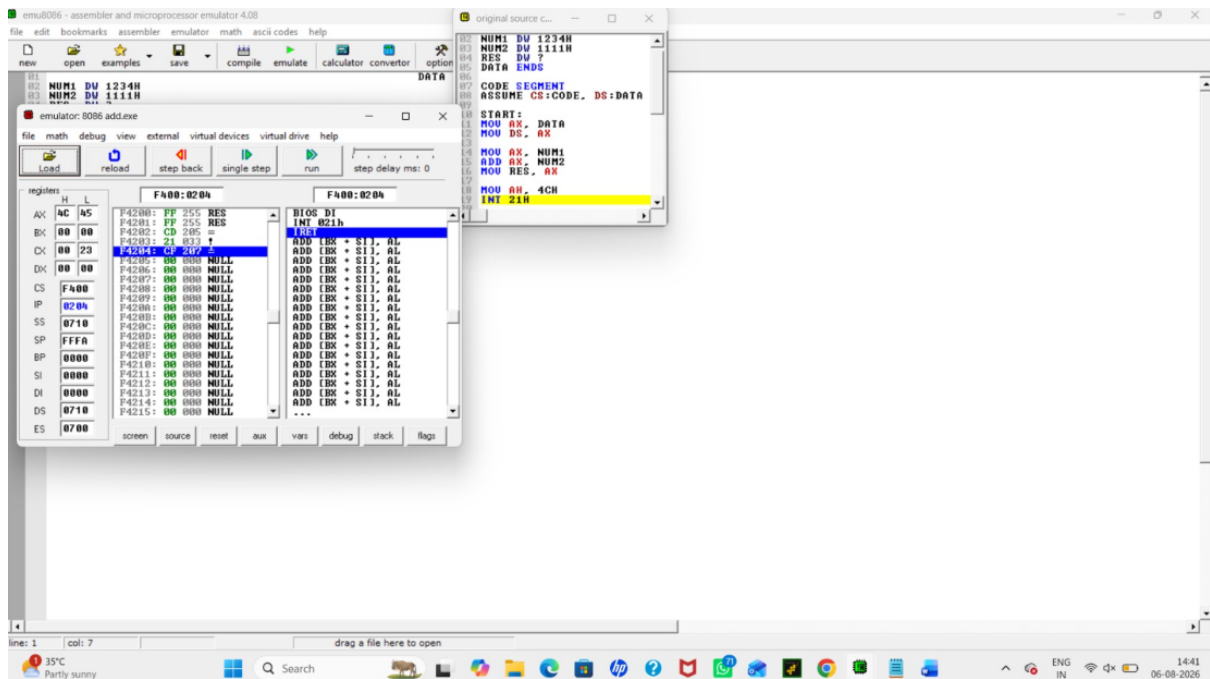
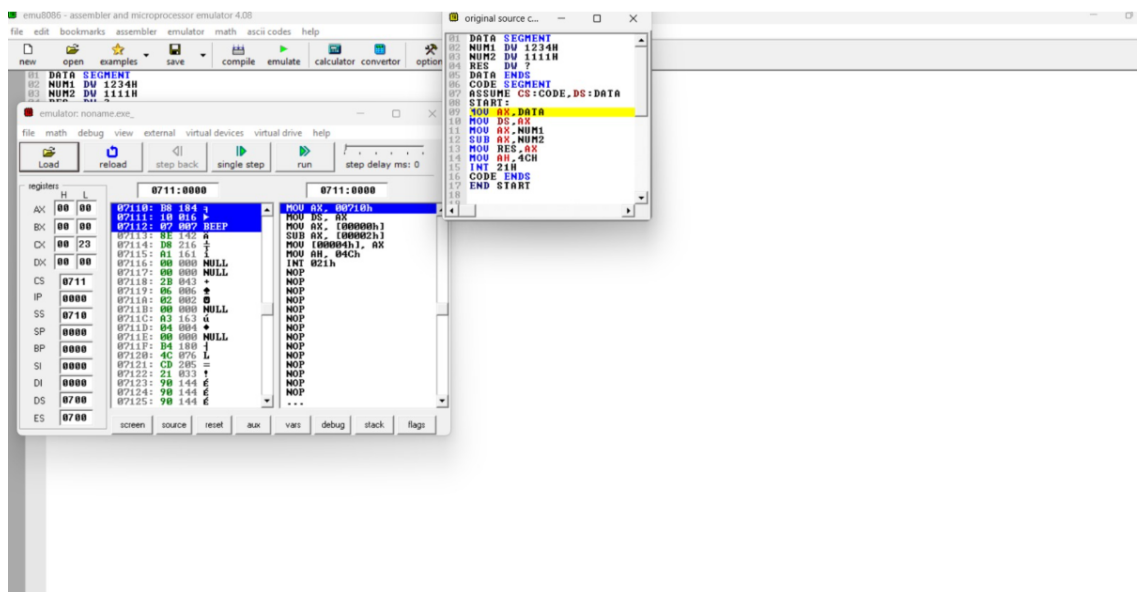


8086 Programs

9. Write an assembly language program for adding two 16-bit data using 8086 processor.



10. Write an assembly language program for subtracting two 16-bit data using 8086 processor.



11. Write an assembly language program for multiplying two 16-bit data using 8086 processor.

The screenshot shows the emu8086 interface with the following assembly code in the 'original source' window:

```

001 DATA SEGMENT
002 NUM1 DW 0005H
003 NUM2 DW 0004H
004 RES DW ?
005 DATA ENDS
006
007 CODE SEGMENT
008 ASSUME CS:CODE, DS:DATA
009
010 START:
011 MOV AX, DATA
012 MOV DS, AX
013
014 MOV AX, NUM1
015 MOV NUM2, AX
016 MOV RES, AX
017 MOV AH, 4Ch
018 INT 21h
019

```

The registers window shows the state of the 8086 registers. The AX register contains 0005H, and the DS register contains 0710H. The memory window shows the data segment starting at 0711:0000.

12. Write an assembly language program for dividing two 16-bit data using 8086 processor.

The screenshot shows the emu8086 interface with the following assembly code in the 'original source' window:

```

001 DATA SEGMENT
002 NUM1 DW 0014H
003 NUM2 DW 0004H
004 QUO DW ?
005 RES DW ?
006 DATA ENDS
007
008 CODE SEGMENT
009 ASSUME CS:CODE, DS:DATA
010
011 START:
012 MOV DS, AX
013
014 MOV AX, NUM1
015 MOV DX, 0000H
016 DIV NUM2
017 MOV QUO, AX
018 MOV RES, AX
019 MOV AH, 4Ch
020 INT 21h
021

```

The registers window shows the state of the 8086 registers. The AX register contains 0014H, and the DS register contains 0710H. The memory window shows the data segment starting at 0711:0000.